

AI-Driven Credit Scoring and Credit Line Solution for the Unreserved and Self-Employed

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Abstract—The financial industry's rapid evolution emphasizes the importance of accurate credit scoring, crucial for decision-making and timely access to loans. Challenges arise due to irregular income streams experienced by self-employed individuals, leading to uncertainty in demonstrating consistent repayment capacity to lenders. Self-employed individuals are often perceived as higher-risk borrowers due to income instability, resulting in stricter lending criteria, higher interest rates, or reduced loan amounts. Customer credit classes are based on default risk, with precise risk assessment vital for profitability. The proposed system employs a Random Forest Classifier Ensemble Learning model on a Two-wheeler Motorcycle Indian Dataset to enhance credit risk assessment comprehensively, especially for self-employed individuals. Credit line predictions incorporating ensemble learning on random forest are being utilized to provide loan amount estimations along with risk considerations, either by taking into account bank statements or other relevant financial data. The proposed system vouches to deliver an accurate result for those who seek aid from financial institutions. The proposed model achieves an accuracy of 87% and good MSE score of 25680.607346699337 by using the Random forest and Ensemble Learning model. The explainable AI graph serves as a valuable tool for transparency and understanding in AI systems, helping users trust and interpret the model's decisions more effectively.

Keywords—AI-based credit score system, Credit line, Ensemble learning, Explainable AI, Loan, Random Forest Classifier, Financial Institution, Bank, Collateral analysis, bank statements.

I. INTRODUCTION

This research introduces an AI-driven credit scoring and credit line system tailored for underserved groups, particularly self-employed and unreserved individuals who often lack conventional credit histories. By leveraging advanced machine learning algorithms and diverse data sources, including factors like cash flow and social connections, this system aims to provide fair and accessible credit assessments. Its holistic approach bridges the credit gap, empowering these individuals to access financial resources for business growth and financial stability. The proposed system utilizes ensemble learning, specifically the Random Forest algorithm, to predict credit scores accurately and efficiently. It also offers risk analysis for banking institutions to facilitate loan decisions, incorporates Explainable AI for transparent spending insights, and designs a loan system based on traditional banking principles.

The research contribution includes:

- Introducing an AI-driven credit scoring system for both employed and self-employed individuals, considering expenditure aspects.
- Providing a credit line solution for risk analysis and loan consideration, tailored for students and self-employed.
- Integrating Explainable AI to offer spending insights and credit score improvement advice.
- Designing a loan system that predicts collateral requirements based on input values.

The study is structured as follows: prior work is discussed in Section 2, followed by a detailed methodology in Section 3, Section 4 focuses light on explainable AI, results and discussion in Section 5, and concluding remarks with future directions in Section 6.

II. RELATED WORK

The literature review for this research encompassed a thorough examination of existing studies on AI-driven credit scoring systems, financial inclusion, and alternative credit assessment methodologies. Various methodologies have been explored, ranging from traditional statistical models to advanced machine learning algorithms.

A. Machine Learning Approaches

Machine learning approaches have significantly impacted credit scoring systems, offering tailored solutions for diverse customer segments. Studies like [1], [2], and [3] have highlighted the effectiveness of machine learning in credit assessment.

For instance, [1] proposed a novel credit approval prediction approach integrating feature selection, instance selection, and decision tree techniques, showcasing improved prediction performance. Similarly, [2] introduced pre-processing and in-processing mitigation methods to mitigate bias in AI systems, while [3] developed a Boosted Tree algorithm for classifying company status and addressing default events. Furthermore, [4] compared various models for borrower default prediction, including logistic regression, random forest, and gradient boosting, emphasizing the application of modern machine learning algorithms and explainable AI tools. Similarly, [5] introduced a stacked classifier approach for efficient credit risk prediction using filter-based feature selection and achieved superior performance over individual estimators. Lastly, [6] proposed leveraging human expertise to compensate for limited data quality, combining machine learning with human-provided rules to enhance decision-making in lending scenarios.

B. Deep Learning based Approach

Deep learning methodologies have revolutionized credit scoring systems, offering enhanced risk assessment capabilities for a wide range of individuals, including business owners, salaried individuals, self-employed individuals, and those with limited credit history. These technologies enable banks to combat fraud, tailor financial services, and make well-informed lending decisions, ultimately improving efficiency and customer satisfaction across diverse demographics. Authors such as Schmitt M [7], Wang C [8], Hayashi Y [9], Diaconescu P [10], Dastile X [11], and others [12] have leveraged deep learning to address

complex challenges and uncover valuable insights in their respective fields. In [7], Schmitt M compares gradient boosting (GB) and deep learning (DL) models for credit scoring, highlighting GB's computational speed advantages and consistency with existing theory and empirical evidence. Similarly, Wang C [8] introduces the FE-Transformer, an end-to-end credit scoring approach utilizing online behavioral data, outperforming traditional methods like logistic regression and XGBoost. Hayashi Y [9] merges a one-dimensional fully connected layer first CNN with the Re-RX algorithm and a decision tree for credit scoring with heterogeneous attributes, prioritizing interpretability and achieving high accuracy. Diaconescu P [10] emphasizes the importance of deep learning in credit scoring, minimizing false negative and false positive errors with optimization algorithms selecting neural network architectures and hyperparameters. Dastile X [11] presents an explainable deep learning model for credit scoring, ensuring compliance with legislation requirements and outperforming traditional methods on publicly available datasets. Additionally, [12] proposes a bidirectional LSTM model for credit card client scoring, demonstrating significant accuracy improvement over traditional machine learning algorithms in predicting missed payments.

C. Hybrid Approach

Financial institutions are increasingly adopting hybrid approaches that blend machine learning and deep learning techniques to calculate credit scores for self-employed individuals. By leveraging machine learning for feature engineering and deep learning for complex pattern recognition, these hybrid methods provide more accurate and dependable credit assessments. Authors such as Munkhdalai L [13], [14], Melese T [15], and others [16] showcase the efficacy of hybrid approaches in addressing complex challenges and gaining insights in their respective domains. Munkhdalai L [13] proposes a hybrid credit scoring model involving neural network models trained in the first phase and merged using logistic regression in the second phase. This model outperforms baseline models on benchmark datasets in terms of H-measure, AUC, and accuracy. Similarly, research [14] introduces a credit scoring system utilizing deep learning and the K-Means algorithm, enhancing accuracy and effectiveness in allocating funds and evaluating customers' creditworthiness.

Melese T [15] presents a hybrid CNN-SVM/RF/DT model for credit-risk prediction, stacking four classifiers to achieve improved prediction performances compared to individual models. Lastly, [16] introduces the Automatic Credit Scoring Strategy (ACSS) to automate credit scoring model construction, integrating modules for data import, model

search, feature selection, and hyperparameter optimization, resulting in notable performance improvements across various credit datasets.

III. PROPOSED SYSTEM

As financial technology advances, big data, and artificial intelligence are gaining prominence in credit risk assessment for enterprises like banks and P2P lending platforms. User default risk is a major concern, prompting firms to turn to deep learning for credit risk assessment. Credit scoring, treated as a classification problem in machine learning, helps categorize applicants as "good" or "bad" customers. This classification aids financial institutions in loan approval and risk pricing decisions [3]. With the shift to online platforms, substantial user behavioral data, including registration, login, clicks, and authentication data, is being generated.

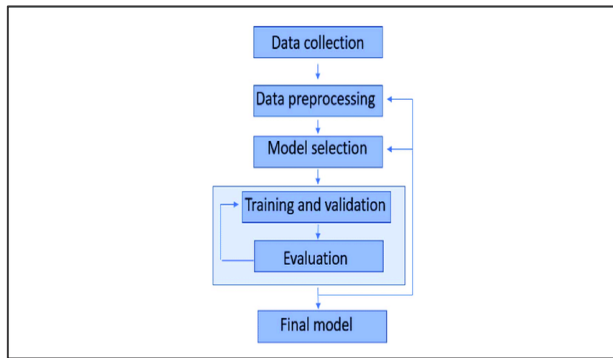


Fig. 1. The proposed framework

Fig. 1 shows the flow diagram of the proposed system and it is discussed below:

1. *Data Collection*: The dataset used in this study focuses on two-wheeler loans obtained by individuals in India [17], containing key parameters such as Age, Gender, Income, Credit Score, and Employment profile. With 279,857 entries, a fraction of the dataset is randomly sampled using the `.sample()` function with `frac=0.1` and `random_state=42` for reproducibility. **Table 1.** presents a subset of the dataset, providing a glimpse into the variables and their corresponding values used in our research.

Table 1. Snap shot of the Dataset

Age	Gender	Income	Credit Score	Employment Profile
31	Male	36000	604	Salaried
25	Male	50000	447	Salaried
62	Other	178000	850	Salaried
69	Female	46000	668	Self-employed

2. *Data Preprocessing*: Data preprocessing is crucial, involving cleaning, transformation, and preparation of raw

data for model training. This includes handling missing values, encoding categorical variables, and scaling numerical features. Categorical data like Gender and Employment Profile is manually encoded into numerical values, ensuring proper formatting for model input.

3. *Feature Selection and Engineering*: Feature selection and engineering involve dividing the data into training and testing sets, tuning hyperparameters for a random forest classifier, and predicting credit scores based on input details like income, employment, age, and marital status. The credit line is determined using predefined formulas, considering factors such as predicted credit score, income, and age (eq. (1)). **Fig. 2.** shows the feature importance graph for our System.

$$\text{credit_line} = \text{predicted_credit_score} * \text{input_income} * (60 - \text{input_age}) * 0.00329065992/10 \quad (1)$$

In eq. (1) 60 is taken as the average retirement age and 0.00329065992/10 is taken as a factor. Every bank has a different formula for calculating the credit score and hence this factor is used. Added to that, we trace back to the earliest transactions recorded and start using the model on the same. Additionally, total income is calculated based on deposits, withdrawals, and balances which is denoted by eq. (2):

$$\text{Total income} = \text{Total Deposits} - \text{Total Withdrawal} + \text{Total Balance} \quad (2)$$

A variable "collateral_number" is derived to assess if collateral is required based on the calculated credit line, determining the variable "collateral_required" as 'yes' or 'no'.

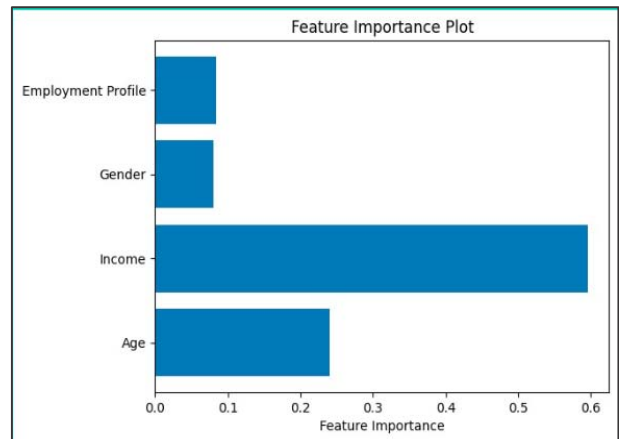


Fig. 2. Feature Importance

4. *Model selection*: In the credit scoring context, multiple models were evaluated, including random forest, Gradient Boosting Machine (GBM), and SVM. The random forest model demonstrated exceptional performance,

achieving an impressive 87% accuracy on both the training and test sets. GBM also yielded a respectable average score of 68% and SVM 74%. A Random forest constructs numerous decision trees during training and outputs the mode of classes for classification or mean prediction for regression. Setting the `random_state` parameter to 42 ensures reproducibility of the results.

5. *Training and Validation:* In the parameter grid, "n_estimators" specifies the number of trees in the forest, with values [50, 100] indicating the options for consideration. "max_depth" determines the maximum depth of each tree, where None implies no depth limit, while a specific integer like 10 sets a maximum depth. "min_samples_split" denotes the minimum number of samples needed to split a node, with options ([2],[5]) for consideration. Similarly, "min_samples_leaf" sets the minimum samples required at a leaf node, with values [1, 2] in the grid. These parameter grids offer a condensed set of combinations for random forest hyperparameters, streamlining the tuning process efficiently. The Random Forest classifier's hyperparameters were fine-tuned using GridSearchCV (Grid Search Cross-Validation). This method exhaustively explores a specified parameter grid, evaluating model performance via cross-validation to determine the best combination of hyperparameters. With a cross-validation fold set to 3, the data is divided into three parts, iteratively training and evaluating the model. The parameter "n_jobs=2" specifies two parallel processes for grid search, enhancing computational efficiency. Additionally, setting "verbose=2" provides detailed output, including progress updates and performance metrics, aiding in model assessment. The final values of the hyperparameters are shown in **Table 2**.

Table 2. Hyperparameters and their values

Hyperparameter	Final Value
n_estimators	100
max_depth	None
min_samples_split	2
min_samples_leaf	1

Ensemble learning utilizes multiple decision trees to improve prediction accuracy. Each tree is trained on a unique subset of data and features, creating diversity within the ensemble. These trees are built through recursive partitioning, where the data is progressively split based on features to create homogeneous sections. This splitting stops at a predefined point, like reaching a certain depth or having a minimum number of data points in each final section. The ensemble's

performance is then assessed using a separate validation dataset to measure metrics like accuracy, providing insights into how well the model performs.

6. *Model Evaluation:* The performance of the ensemble model was evaluated using various metrics, and hyperparameters were tuned to optimize performance. The score method of the Random Forest classifier was used to calculate the accuracy of the model on the test data. This method takes the test features and corresponding true labels as input and returns the accuracy of the model.

7. *Final Model:* The final model for credit score prediction utilized ensemble learning on a Random forest approach combined with a filter-based feature selection method. It generated credit lines based on the credit scores generated by the AI model, ensuring fairness and transparency in the allocation process.

IV. EXPLAINABLE AI

This proposed system utilizes Explainable AI (XAI) to create a trustworthy and transparent credit scoring system for unreserved and self-employed individuals. Traditional AI models can be opaque, making it difficult to understand how they reach decisions. XAI tackles this by providing explanations for the AI's recommendations.

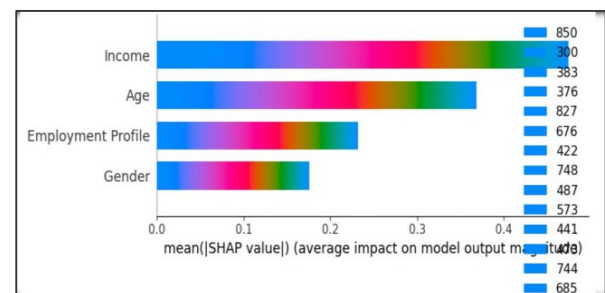


Fig. 3. Explainable AI for a customer

In **Fig. 3**. The explainable AI graph shows the various factors that have been taken into account. **Fig. 3**. shows that the blue areas with respect to the factors show a positive impact. It will take the age into consideration and then accordingly look at the other factors, such as income and employment profile, accordingly suggesting alternate methods. The first step is choosing machine learning algorithms that are inherently interpretable. These include decision trees and rule-based models. These algorithms prioritize transparency, allowing users to understand how their creditworthiness is assessed.

Secondly, the system is designed to generate explanations alongside its predictions. This might involve highlighting features that significantly impact the score or providing reasoning based on decision rules. Finally, users can interact

with the system to explore the rationale behind its recommendations, fostering trust and understanding.

V. RESULTS AND DISCUSSION

To validate the proposed model, we used **PaisaBazaar.com** as our benchmark to compare the scores we have obtained.

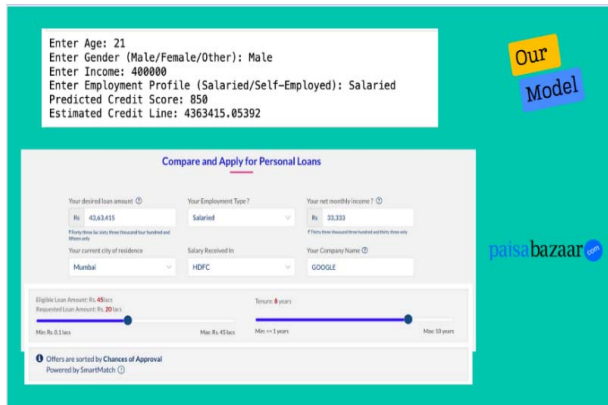


Fig. 4. Results of our system after comparison with present systems

As shown in **Fig 4**, PaisaBazaar.com is used as our benchmark to compare the scores we have obtained. As you can see in the image above, we have been able to match the model used by the website, albeit with some differences. We can also note that for the same set of values, we are getting a pretty similar predicted credit score. Whilst comparing, we have also found out that PaisaBazaar.com has not been able to provide a curve for the predicted score. By providing the end consumer with a graphical representation as to how much of a loan amount will be safe, we will be able to give the consumer a good idea of how they need to manage their spends and also bring about a change in their spending habits, which will be suggested by the explainable AI.

Model comparisons:

Before zeroing down on ensemble learning, we had tried gradient boosting as an alternative, the results of which were not as promising as the current ones. The factors mentioned above - income, employment, age, gender- all were taken into account and there was no change with respect to any values whatsoever. However, it is the overall accuracy of the model that prompted us for a change. For the similar dataset, when run with the same model, the end accuracy result was 68.1 percent. In a market where accurate results will be of the utmost importance, a value of this parameter will be deemed as insufficient and could thus doubt the credibility of the model.

Below mentioned is the matrix and the overall values we achieved when tested with Gradient Boosting:

Table 3 . Comparison between ML models

Method	Accuracy	R-Squared
Random Forest	87.17%	0.0697359170
XGBoost	65%	0.2987654345
SVM	74%	0.097656655

As shown in Table 3. out of the three models, the Random forest performs better. We got a positive and affirmative accuracy percentage for the random forest model. It helped us in getting much more accurate results and showed a rating of 87 percent accuracy.

Evaluation metrics: Evaluation metrics used for comparing various proposed models are: Accuracy and MSE. Using the following testing metrics, we have calculated how the model will be able to calculate the values present in the dataset and will also be able to ensure that the accuracy is good enough along with the prescribed metrics we have followed.

As shown in **Fig. 5**, we have been able to achieve an accuracy of 87% and a pretty good MSE score of 25680.607346699337.

```
Mean Squared Error: 25680.607346699337
Enter Age: 25
Enter Gender (Male/Female/Other): Male
Enter Income: 30000
Enter Employment Profile (Salaried/Self-Employed): Self Employed
Predicted Credit Score: 626.8125
Credit Line: 2165758.1096602497
```

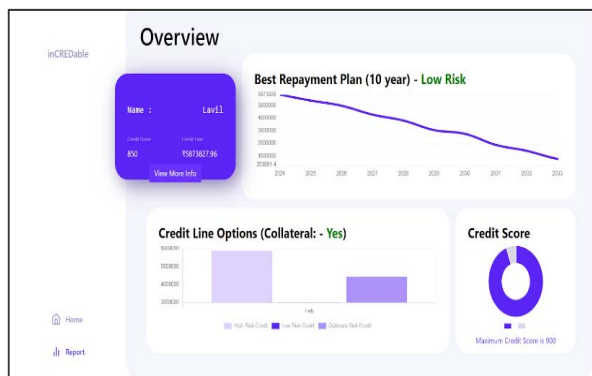
Fig 5. Results of the proposed system

Graphical User Interface:

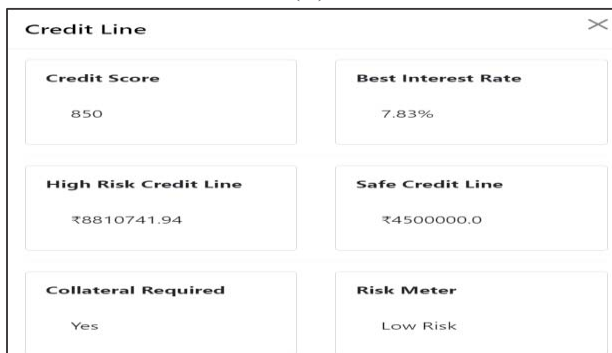
We developed a user-friendly website for the users to use the proposed AI based credit score prediction system.

Landing and form Page: Fig 6. shows the landing page of the website. Here, the user will be asked to fill in their details for the calculation of the credit score. The metrics will be taken into account and then it will be calculated. By doing so, the values will be noted down and then the necessary fields (age, gender, and income) will be taken into account for the next step.

Fig 6. GUI of our proposed system



(A)



(B)

Fig. 7. Report Overview page and the Credit Line explanations

Report Page: Fig 7 shows the report page of the AI based credit score website. This is the web page where the graphs show the necessary information after calculating all the metrics. Name, age, credit score, and the spending-to-earning ratio are displayed in the form of graphs and explained accordingly. The loan collateral feature indicates how much of a loan amount will be enough for the individual to apply based on the balance sheet provided. Once calculated, it gives a high-risk, low-risk, and average-risk credit line. The user can also analyze the same and take a safe loan amount accordingly

All of the above results will be powered by our Explainable AI system. In this system, inputs after carrying out all the

calculations are considered and thus will be also able to prepare a spending plan for the end user. There will be specially curated plans for the same and the end user can then go ahead and follow the plan set up to achieve either one of the credit line amounts shortlisted by the AI.

VI CONCLUSION

This research mainly focuses on providing credit score and credit line, especially for unreserved and self-employed individuals. The research work employed an ensemble learning method using the random forest to predict the credit score. Using the predicted credit score the study also provides the credit line to the end users with the possible risk analysis. Also, the proposed system incorporates the Explainable AI to create a trustworthy and transparent credit scoring system for unreserved and self-employed individuals. The system, if implemented, would help banks cater to the unreserved people who fail to have access to basic loan facilities. In the future, this system will help MSMEs (Micro, Small, and Medium Enterprises) to grow and expand their ventures into other parts that will help their growth. This system will also help the students, who are looking for loans to help with their postgraduate studies and will aid them in getting loans hassle-free.

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